C6 Equilibria – Knowledge Organiser

AQA A-level Chemistry • Set 1: Dynamic Equilibrium and Le Chatelier **Papers 1 & 2**

#	Question	Answer
1	Dynamic equilibrium	Forward and reverse rates equal; concentrations constant
2	Condition needed to reach equilibrium	Closed system
3	Symbol for a reversible reaction	⇌
4	Le Chatelier's principle	Equilibrium shifts to oppose a change
5	Effect of a catalyst on equilibrium position	None
6	Effect of a catalyst on time to reach equilibrium	Shorter
7	Homogeneous equilibrium	All species in the same phase
8	Two disadvantages of high pressure	Expensive and hazardous
9	Equation for hydration of ethene	$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$
10	Catalyst for hydration of ethene	Phosphoric acid (H_3PO_4)

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AQA A-level Chemistry • Set 2: Equilibrium Constant K_c **Papers 1 & 2**

#	Question	Answer
1	K_c for $aA + bB \rightleftharpoons cC + dD$	$[C]^c[D]^d \div [A]^a[B]^b$
2	Meaning of square brackets	Concentration in mol dm^{-3}
3	Only factor that changes K_c	Temperature
4	Effect of concentration on K_c	None
5	Effect of pressure on K_c	None
6	Effect of a catalyst on K_c	None
7	K_c much greater than 1	Equilibrium lies to the right
8	K_c much less than 1	Equilibrium lies to the left
9	Equation for methanol production	$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$
10	Conditions for hydration of ethene	300 °C, 6 MPa